

- b. The Galerkin solution is exact at the endpoints of each subinterval for all cases.
- c. In each case, there is at least one point in each subinterval at which u' is exact.

After generalizing to the case of n subintervals in the following section, we shall show in Sec. 1.10 that the above observations are not accidental.

Exercise 1. If the reader has not had experience with calculations of the type presented in this section, it would be worthwhile to reproduce all results, providing all omitted details.

1.8 PIECEWISE LINEAR FINITE ELEMENT SPACE

The examples of the preceding section employed definitions of $\mathcal{V}^h$ and $\mathcal{S}^h$ which were special cases of the so-called piecewise linear finite element space. To define the general case in which $\mathcal{V}^h$ is n -dimensional, we partition the domain $[0, 1]$ into n nonoverlapping subintervals. The typical subinterval is denoted by $[x_A, x_{A+1}]$, where $x_A < x_{A+1}$ and $A = 1, 2, \dots, n$. We also require $x_1 = 0$ and $x_{n+1} = 1$. The x_A 's are called **nodal points**, or simply **nodes**. (The terminologies *joints* and *knots* are also used.) The subintervals are sometimes referred to as the **finite element domains**, or simply **elements**. Notice that the lengths of the elements, $h_A = x_{A+1} - x_A$, are *not* required to be equal. The mesh parameter, h , is generally taken to be the length of the maximum subinterval (i.e., $h = \max h_A, A = 1, 2, \dots, n$). The smaller h , the more "refined" is the partition, or mesh. If the subinterval lengths are equal, then $h = 1/n$.

The shape functions are defined as follows: Associated to a typical internal node (i.e., $2 \leq A \leq n$)

$$N_A(x) = \begin{cases} \frac{(x - x_{A-1})}{h_{A-1}}, & x_{A-1} \leq x \leq x_A \\ \frac{(x_{A+1} - x)}{h_A}, & x_A \leq x \leq x_{A+1} \\ 0, & \text{elsewhere} \end{cases} \quad (1.8.1)$$

whereas for the boundary nodes we have

$$N_1(x) = \frac{x_2 - x}{h_1}, \quad x_1 \leq x \leq x_2 \quad (1.8.2)$$

$$N_{n+1}(x) = \frac{x - x_n}{h_n}, \quad x_n \leq x \leq x_{n+1} \quad (1.8.3)$$

The shape functions are sketched in Fig. 1.8.1. For obvious reasons, they are referred to variously as "hat," "chapeau," and "roof" functions. Note that $N_A(x_B) = \delta_{AB}$, where

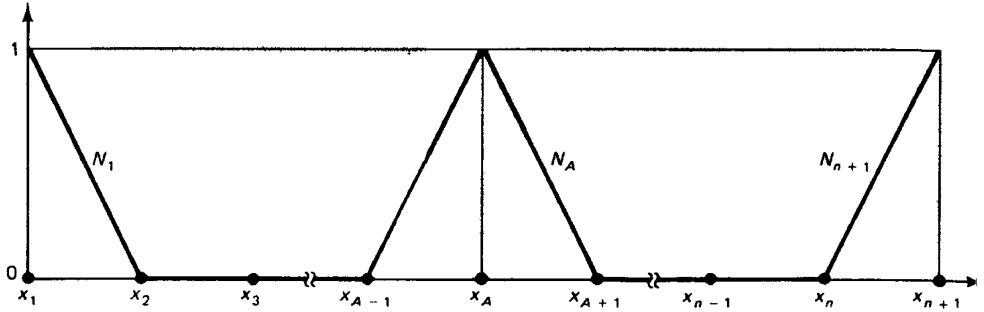
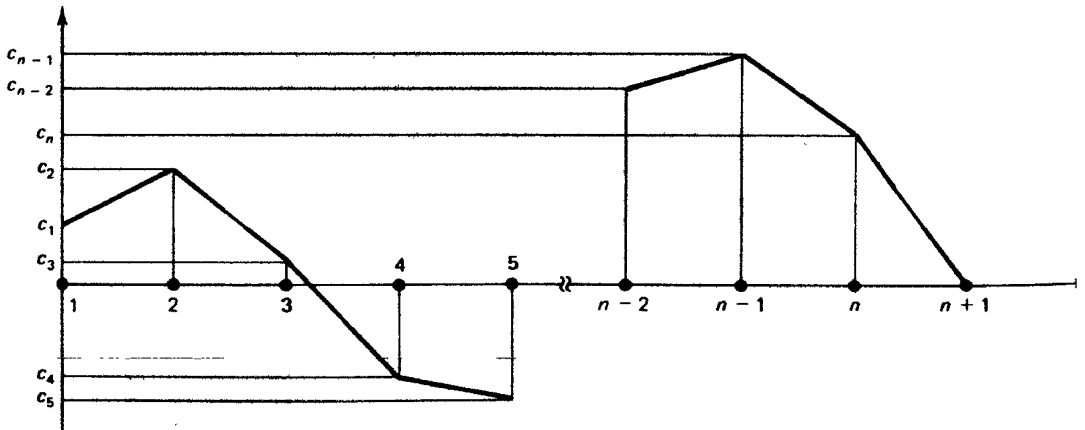


Figure 1.8.1 Basis functions for the piecewise linear finite element space.

δ_{AB} is the Kronecker delta (i.e., $\delta_{AB} = 1$ if $A = B$, whereas $\delta_{AB} = 0$ if $A \neq B$). In words, N_A takes on the value 1 at node A and is 0 at all other nodes. Furthermore, N_A is nonzero only in the subintervals that contain x_A .

A typical member $w^h \in \mathcal{U}^h$ has the form $\sum_{A=1}^n c_A N_A$ and appears as in Fig. 1.8.2. Note that w^h is continuous but has discontinuous slope across each element boundary. For this reason, w_x^h , the generalized derivative of w^h , will be piecewise constant, experiencing discontinuities across element boundaries. (Such a function is sometimes called a *generalized step function*.) Restricted to each element domain, w^h is a linear polynomial in x . In respect to the homogeneous essential boundary condition, $w^h(1) = 0$. Clearly, w^h is identically zero if and only if each $c_A = 0$, $A = 1, 2, \dots, n$.


 Figure 1.8.2 A typical member $w^h \in \mathcal{U}^h$.

Typical members of $\mathcal{S}^h$ are obtained by adding $q^h = q N_{n+1}$ to typical members of $\mathcal{U}^h$. This ensures that $u^h(1) = q$.

The piecewise linear finite element functions are the simplest and most widely used finite element functions for one-dimensional problems.

Exercise 1. Consider the weak formulation of the one-dimensional model problem:

$$\int_0^1 w_{,x} u_{,x} dx = \int_0^1 w f dx + w(0)h \quad (1.8.4)$$

where $w \in \mathcal{U}$ and $u \in \mathcal{S}$ are assumed to be smooth on element interiors (i.e., on $]x_A, x_{A+1}[$, $A = 1, 2, \dots, n$), but may suffer slope discontinuities across element boundaries. (Functions of this class contain the piecewise linear finite element space described earlier.) From (1.8.4) and the assumed continuity of the functions, show that:

$$\begin{aligned} 0 = & \sum_{A=1}^n \int_{x_A}^{x_{A+1}} w(u_{,xx} + f) dx + w(0)[u_{,x}(0^+) + h] \\ & + \sum_{A=2}^n w(x_A)[u_{,x}(x_A^+) - u_{,x}(x_A^-)] \end{aligned} \quad (1.8.5)$$

Arguing as in Sec. 1.4, it may be concluded that the Euler-Lagrange conditions of (1.8.5) are

- i. $u_{,xx}(x) + f(x) = 0$, where $x \in]x_A, x_{A+1}[$ and $A = 1, 2, \dots, n$,
- ii. $-u_{,x}(0^+) = h$; and
- iii. $u_{,x}(x_A^-) = u_{,x}(x_A^+)$, where $A = 2, 3, \dots, n$.

Observe that (i) is the differential equation *restricted to element interiors*, and (iii) is a continuity condition across element boundaries. This may be contrasted with the case in which the solution is assumed smooth. In this case the continuity condition is identically satisfied and the summation of integrals over element interiors may be replaced by an integral over the entire domain (see Sec. 1.4).

In the Galerkin finite element formulation, an *approximate* solution of (i)–(iii) is obtained.

1.9 PROPERTIES OF K

The shape functions N_A , $A = 1, 2, \dots, n+1$, are zero outside a neighborhood of node A . As a result, many of the entries of K are zero. This can be seen as follows. Let $B > A+1$. Then (see Fig. 1.9.1)

$$K_{AB} = \int_0^1 \underbrace{N_{A,x} N_{B,x}}_0 dx = 0 \quad (1.9.1)$$

The symmetry of K implies, in addition, that (1.9.1) holds for $A > B+1$. One says that K is **banded** (i.e., its nonzero entries are located in a band about the main diagonal). Figure 1.9.2 depicts this property. Banded matrices have significant advantages in that the zero elements outside the band neither have to be stored nor operated

Sec. 1.9 Properties of K

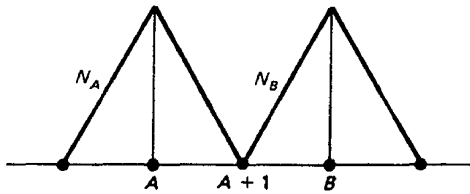


Figure 1.9.1 If $B > A + 1$, the zero portions of N_B and N_A do not overlap.

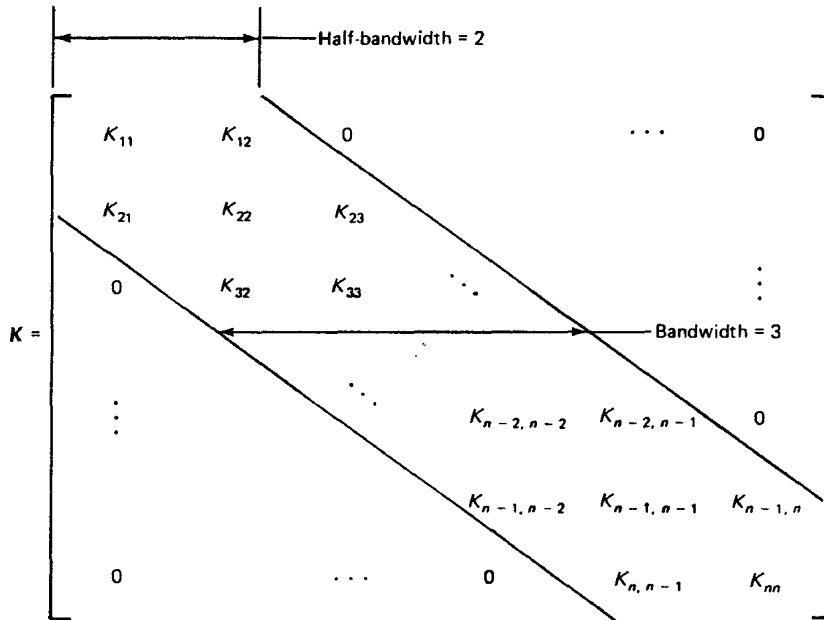


Figure 1.9.2 Band structure of K .

upon in the computer. The stiffness matrix arising in finite element analysis is, general, narrowly banded, lending itself to economical formation and solution.

Definition. An $n \times n$ matrix A is said to be *positive definite* if

- i. $c^T A c \geq 0$ for all n -vectors c ; and
- ii. $c^T A c = 0$ implies $c = 0$.

Remarks

1. A symmetric positive-definite matrix possesses a unique inverse.
2. The eigenvalues of a positive-definite matrix are real and positive.

Theorem. The $n \times n$ matrix K defined by (1.6.11) is positive definite.

Proof

- i. Let c_A , $A = 1, 2, \dots, n$, be the components of c (i.e., $c = \{c_A\}$), arbitrary vector. Use these c_A 's to construct a member of $\mathcal{U}^h$, $w^h = \sum_{A=1}^n c_A N_A$, wh

the N_A 's are the basis functions for $\mathcal{U}^h$. Then

$$\begin{aligned}
 \mathbf{c}^T \mathbf{K} \mathbf{c} &= \sum_{A,B=1}^n c_A K_{AB} c_B \\
 &= \sum_{A,B=1}^n c_A a(N_A, N_B) c_B && \text{(definition of } K_{AB}) \\
 &= a\left(\sum_{A=1}^n c_A N_A, \sum_{B=1}^n c_B N_B\right) && \text{(bilinearity of } a(\cdot, \cdot)) \\
 &= a(w^h, w^h) && \text{(definition of } w^h) \\
 &= \int_0^1 \underbrace{(w_{,x}^h)^2}_{\geq 0} dx && \text{(by (1.4.8))} \\
 &\geq 0
 \end{aligned}$$

ii. Assume $\mathbf{c}^T \mathbf{K} \mathbf{c} = 0$. By the proof of part (i),

$$\int_0^1 (w_{,x}^h)^2 dx = 0$$

and consequently w^h must be constant. Since $w^h \in \mathcal{U}^h$, $w^h(1) = 0$. Combining these facts, we conclude that $w^h(x) = 0$ for all $x \in [0, 1]$, which is possible only if each $c_A = 0$, $A = 1, 2, \dots, n$. Thus $\mathbf{c} = \mathbf{0}$. ■

Note that part (ii) depended upon the definition of $\mathbf{K}$ and the zero essential boundary condition built into the definition of $\mathcal{U}^h$.

Summary. $\mathbf{K}$, defined by (1.6.11), is

- i. Symmetric
- ii. Banded
- iii. Positive-definite

The practical consequence of the above properties is that a very efficient computer solution of $\mathbf{K} \mathbf{d} = \mathbf{F}$ may be performed.

1.10 MATHEMATICAL ANALYSIS

In this section we will show that the observations made with reference to the example problems of Sec. 1.7 are, in fact, general results. To establish these facts rigorously requires only elementary mathematical techniques.

Our first objective is to establish that the Galerkin finite element solution u^h is exact at the nodes. To do this we must introduce the notion of a Green's function.

Let $\delta_y(x) = \delta(x - y)$ denote the *Dirac delta function*. The Dirac function is not a function in the classical sense but rather an operator defined by its action on

1.12 THE ELEMENT POINT OF VIEW

So far we have viewed the finite element method simply as a particular Galerkin approximation procedure applied to the weak statement of the problem in question. What makes what we have done a finite element procedure is the character of the selected basis functions; particularly their piecewise smoothness and "local support" (i.e., $N_A \equiv 0$ outside a neighborhood of node A). This is the mathematical point of view; it is a **global** point of view in that the basis functions are considered to be defined everywhere on the domain of the boundary-value problem. The global viewpoint is useful in establishing the mathematical properties of the finite element method. This can be seen in Sec. 1.10 and will be made more apparent later on.

Now we wish to discuss another point of view called the **local**, or **element**, point of view. This viewpoint is the traditional one in engineering and is useful in the computer implementation of the finite element method and in the development of finite elements.

We begin our treatment of the local point of view with a question: What is a finite element?

We shall attempt to give the answer in terms of the piecewise linear finite element space that we defined previously. An individual element consists of the following quantities.

Linear finite element (global description)

- (g 1) Domain: $[x_A, x_{A+1}]$
- (g 2) Nodes: $\{x_A, x_{A+1}\}$
- (g 3) Degrees of freedom: $\{d_A, d_{A+1}\}$
- (g 4) Shape functions:⁵ $\{N_A, N_{A+1}\}$
- (g 5) Interpolation function:

$$u^h(x) = N_A(x)d_A + N_{A+1}(x)d_{A+1}, \quad x \in [x_A, x_{A+1}]$$

(Recall $d_A = u^h(x_A)$.) In words, a linear finite element is just the totality of paraphernalia associated with the globally defined function u^h *restricted* to the *element* domain. The above quantities are in terms of *global* parameters—namely, the global coordinates, global shape functions, global node ordering, and so on. It is fruitful to introduce a *local* set of quantities, corresponding to the global ones, so that calculations for a typical element may be standardized. These are given as follows:

Linear finite element (local description)

- (l 1) Domain: $[\xi_1, \xi_2]$

⁵ In weighted residual methods in which $\mathcal{S}^h$ and $\mathcal{U}^h$ are built up from different classes of functions (i.e., Petrov-Galerkin methods), we would also have to specify a set of **weighting functions**, say $\{\tilde{N}_A, \tilde{N}_{A+1}\}$; the entire set of $\tilde{N}_A$'s would then constitute a basis for $\mathcal{U}^h$. In Galerkin's method $\tilde{N}_A = N_A$.

$$(12) \quad \text{Nodes:} \quad \{\xi_1, \xi_2\}$$

$$(13) \quad \text{Degrees of freedom:} \quad \{d_1, d_2\}$$

$$(14) \quad \text{Shape functions:} \quad \{N_1, N_2\}$$

$$(15) \quad \text{Interpolation function:}$$

$$u^h(\xi) = N_1(\xi)d_1 + N_2(\xi)d_2$$

Note that in the local description, the nodal numbering begins with 1.

We shall relate the domains of the global and local descriptions by an “affine” transformation $\xi : [x_A, x_{A+1}] \rightarrow [\xi_1, \xi_2]$, such that $\xi(x_A) = \xi_1$ and $\xi(x_{A+1}) = \xi_2$. It is standard practice to take $\xi_1 = -1$ and $\xi_2 = +1$. Thus ξ may be represented by the expression

$$\xi(x) = c_1 + c_2 x \quad (1.12.1)$$

where c_1 and c_2 are constants which are determined by

$$\left. \begin{aligned} -1 &= c_1 + x_A c_2 \\ 1 &= c_1 + x_{A+1} c_2 \end{aligned} \right\} \quad (1.12.2)$$

Solving this system yields

$$\xi(x) = \frac{2x - x_A - x_{A+1}}{h_A} \quad (1.12.3)$$

(Recall $h_A = x_{A+1} - x_A$.) The inverse of ξ is obtained by solving for x :

$$x(\xi) = \frac{h_A \xi + x_A + x_{A+1}}{2} \quad (1.12.4)$$

In (1.12.1), ξ is a mapping and x is a point, whereas in (1.12.4), x is a mapping and ξ is a point.

In the sequel, we adopt the notational convention that subscripts $a, b, c, \dots$ pertain to the local numbering system. The subscripts $A, B, C, \dots$ will always pertain to the global numbering system. To control the proliferation of notations, we will frequently use the same notation for the local and global systems (e.g., d_a and d_A or N_a and N_A). This generally should not cause confusion as the context will make clear which point of view is being adopted. If there is danger of confusion, a superscript e will be introduced to denote a quantity in the local description associated with element number e (e.g., $d_a^e = d_A$, $N_a^e(\xi) = N_A(x^e(\xi))$, where $x^e : [\xi_1, \xi_2] \rightarrow [x_1^e, x_2^e] = [x_A, x_{A+1}]$, etc.).

In terms of ξ , the shape functions in the local description take on a standard form

$$N_a(\xi) = \frac{1}{2}(1 + \xi_a \xi), \quad a = 1, 2 \quad (1.12.5)$$

Note also that (1.12.4) may be written in terms of (1.12.5):

$$x^e(\xi) = \sum_{a=1}^2 N_a(\xi) x_a^e. \quad (1.12.6)$$

This has the same form as the interpolation function (cf. 15).

For future reference, we note the following results:

$$N_{a,\xi} = \frac{\xi_a}{2} = \frac{(-1)^a}{2} \quad (1.12.7)$$

$$x_{,\xi}^e = \frac{h^e}{2} \quad (1.12.8)$$

where $h^e = x_2^e - x_1^e$ and

$$\xi_{,x}^e = (x_{,\xi}^e)^{-1} = \frac{2}{h^e} \quad (1.12.9)$$

The local and global descriptions of the e th element are depicted in Fig. 1.12.1.

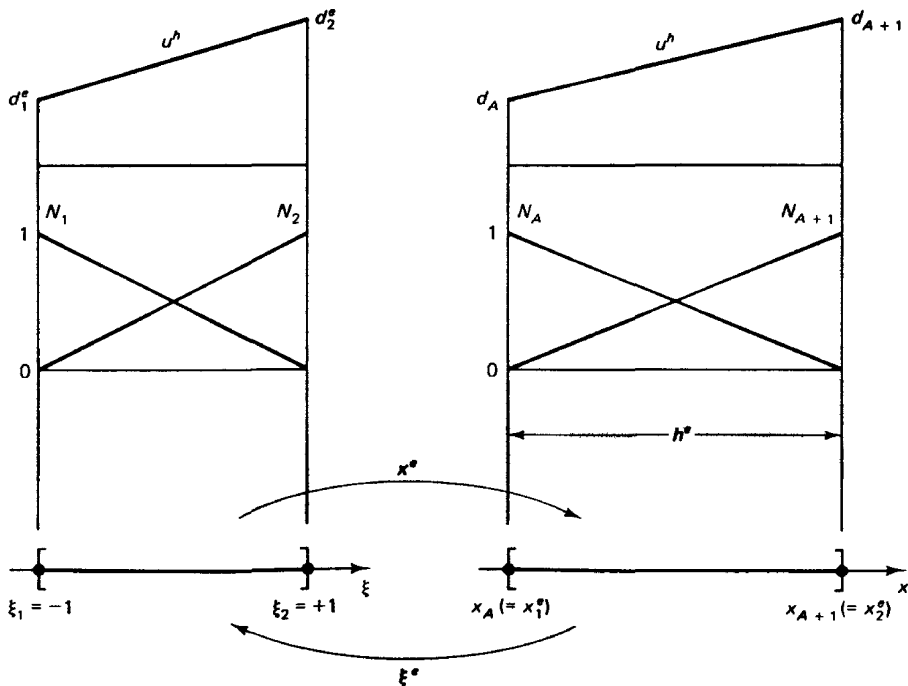


Figure 1.12.1 Local and global descriptions of the e th element.

1.13 ELEMENT STIFFNESS MATRIX AND FORCE VECTOR

To develop the element point of view further, let us assume that our model consists of n_{el} elements, numbered as shown in Figure 1.13.1. Clearly $n_{el} = n$ for this case. Let us take e to be the variable index for the elements; thus $1 \leq e \leq n_{el}$.

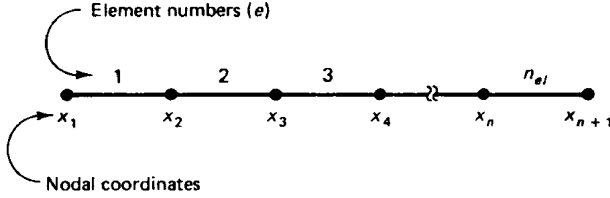


Figure 1.13.1

Now recall the definitions of the (global) stiffness matrix and force vector

$$K = \underbrace{[K_{AB}]}_{n \times n}, \quad F = \underbrace{\{F_A\}}_{n \times 1} \quad (1.13.1)$$

where

$$K_{AB} = a(N_A, N_B) = \int_0^1 N_{A,x} N_{B,x} dx \quad (1.13.2)$$

$$\begin{aligned} F_A &= (N_A, \ell) + \delta_{A1} h - a(N_A, N_{n+1})q \\ &= \int_0^1 N_A \ell dx + \delta_{A1} h - \int_0^1 N_{A,x} N_{n+1,x} dx q \end{aligned} \quad (1.13.3)$$

(In (1.13.3) we have assumed $N_A(x_1) = \delta_{A1}$, as for the piecewise linear finite element space.) The integrals over $[0, 1]$ may be written as sums of integrals over the element domains. Thus

$$K = \sum_{e=1}^{n_{el}} K^e, \quad K^e = [K_{AB}^e] \quad (1.13.4)$$

$$F = \sum_{e=1}^{n_{el}} F^e, \quad F^e = \{F_A^e\} \quad (1.13.5)$$

where

$$K_{AB}^e = a(N_A, N_B)^e = \int_{\Omega^e} N_{A,x} N_{B,x} dx \quad (1.13.6)$$

$$\begin{aligned} F_A^e &= (N_A, \ell)^e + \delta_{e1} \delta_{A1} h - a(N_A, N_{n+1})^e q \\ &= \int_{\Omega^e} N_A \ell dx + \delta_{e1} \delta_{A1} h - \int_{\Omega^e} N_{A,x} N_{n+1,x} dx q \end{aligned} \quad (1.13.7)$$

and $\Omega^e = [x_1^e, x_2^e]$, the domain of the e th element.

The important observation to make is that K and F can be constructed by

summing the contributions of elemental matrices and vectors, respectively. In the literature, this procedure is sometimes called the *direct stiffness method* [10].

By the definitions of the N_A 's, we have that

$$K_{AB}^e = 0, \quad \text{if } A \neq e \text{ or } e + 1 \text{ or } B \neq e \text{ or } e + 1 \quad (1.13.8)$$

and

$$F_A^e = 0, \quad \text{if } A \neq e \text{ or } e + 1 \quad (1.13.9)$$

The situation for a typical element, e , is shown in Fig. 1.13.2. In practice we would not, of course, add in the zeros but merely add in the nonzero terms to the appropriate locations. For this purpose it is useful to define the *eth element stiffness matrix* k^e and *element force vector* f^e as follows:

$$k^e = \underbrace{[k_{ab}^e]}_{2 \times 2}, \quad f^e = \underbrace{\{f_a^e\}}_{2 \times 1} \quad (1.13.10)$$

$$k_{ab}^e = a(N_a, N_b)^e = \int_{\Omega^e} N_{a,x} N_{b,x} dx \quad (1.13.11)$$

$$f_a^e = \int_{\Omega^e} N_a f dx + \begin{cases} \delta_{a1} h & e = 1 \\ 0 & e = 2, 3, \dots, n_{el} - 1 \\ -k_{a2}^e q & e = n_{el} \end{cases} \quad (1.13.12)$$

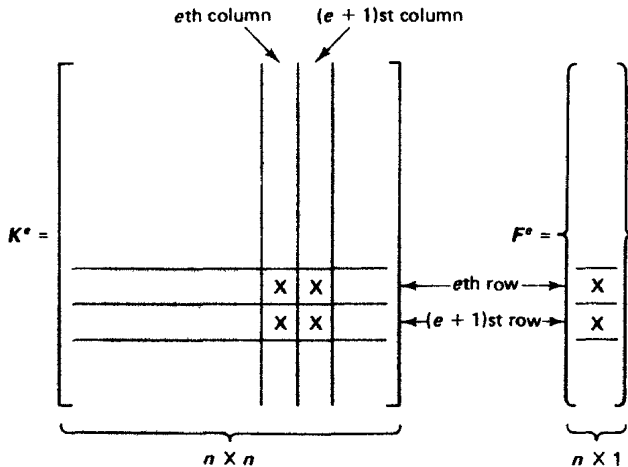


Figure 1.13.2 X's indicate nonzero terms; all other terms are zero.

Here k^e and f^e are defined with respect to the *local* ordering, whereas K^e and F^e are defined with respect to the *global* ordering. To determine where the components of k^e and f^e "go" in K and F , respectively, requires keeping additional information. This is discussed in the following section.

1.14 ASSEMBLY OF GLOBAL STIFFNESS MATRIX AND FORCE VECTOR; LM ARRAY

In a finite element computer program, it is the task of a “finite element subroutine” to produce k^e and f^e , $e = 1, 2, \dots, n_{el}$, from given data and to provide an “assembly subroutine” enough information so that the terms in k^e and f^e can be added to the appropriate locations in K and F , respectively. This assembly information is stored in an array named LM, the *location matrix*.

Let us construct the LM array for the problem under consideration. The dimensions of LM are n_{en} , the *number of element nodes*, by the number of elements; in the present case, the numbers are 2 and n_{el} , respectively. Given a particular degree of freedom number and an element number (say a and e , respectively), the value returned by the LM array is the corresponding global equation number, A , viz.,

$$A = \text{LM}(a, e) = \begin{cases} e & \text{if } a = 1 \\ e + 1 & \text{if } a = 2 \end{cases} \quad (1.14.1)$$

The complete LM array is depicted in Fig. 1.14.1. This is the way we envision it stored in the computer. Note that $\text{LM}(2, n_{el}) = 0$. This indicates that degree of freedom 2 of element number n_{el} is prescribed and is not an unknown in the global matrix equation. Hence the terms $k_{12}^{n_{el}}$, $k_{21}^{n_{el}}$, $k_{22}^{n_{el}}$, and $f_2^{n_{el}}$ are *not* assembled into K and F , respectively. (There are no places for them to go!)

		Element numbers $1 \leq e \leq n_{el}$							
		1	2	3	...	e	...	$n_{el}-1$	n_{el}
Local node number	1	1	2	3	...	e	...	$n-1$	n
	2	2	3	4	...	$e+1$	...	n	0
$(n_{en} = 2)$		$n_{en} \times n_{el}$							

Figure 1.14.1 LM array for example problem.

As an example, assume we want to add the e th elemental contributions, where $1 \leq e \leq n_{el}-1$, to the partially assembled K and F . From the LM array, we deduce the following assembly procedure:

$$K_{ee} \leftarrow K_{ee} + k_{11}^e \quad (1.14.2)$$

$$K_{e,e+1} \leftarrow K_{e,e+1} + k_{12}^e \quad (1.14.3)$$

$$K_{e+1,e} \leftarrow K_{e+1,e} + k_{21}^e \quad (1.14.4)$$

$$K_{e+1,e+1} \leftarrow K_{e+1,e+1} + k_{22}^e \quad (1.14.5)$$

⁶ Due to symmetry k_{21}^e would not actually be assembled in practice.

$$F_e \leftarrow F_e + f_1^e \quad (1.14.6)$$

$$F_{e+1} \leftarrow F_{e+1} + f_2^e \quad (1.14.7)$$

where the arrow ($\leftarrow$) is read “is replaced by.”

For element n_{el} we have only that

$$K_{nn} \leftarrow K_{nn} + k_{11}^{n_{el}} \quad (1.14.8)$$

$$F_n \leftarrow F_n + f_1^{n_{el}} \quad (1.14.9)$$

With these ideas, we may construct, in sketchy fashion, an algorithm for the assembly of K and F ; see Fig. 1.14.2.

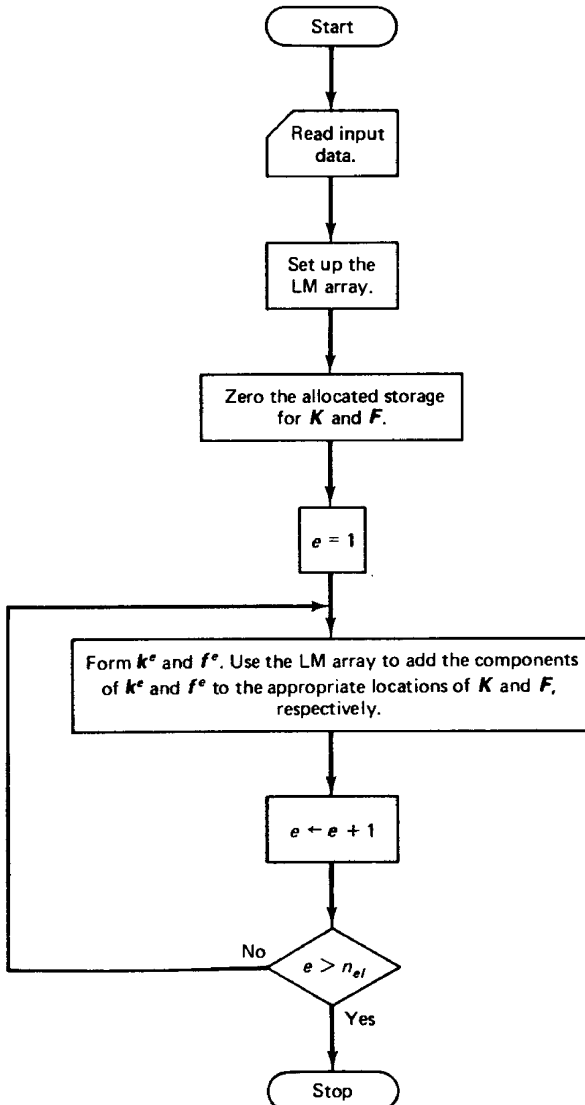


Figure 1.14.2 Flowchart of a finite element assembly algorithm.

The action of the assembly algorithm is denoted throughout by $\mathbf{A}$, the *assembly operator*, vis.,

$$\mathbf{K} = \mathbf{A}_{e=1}^{n_{el}} (k^e), \quad \mathbf{F} = \mathbf{A}_{e=1}^{n_{el}} (f^e) \quad (1.14.10)$$

1.15 EXPLICIT COMPUTATION OF ELEMENT STIFFNESS MATRIX AND FORCE VECTOR

The explicit computation of k^e and f^e , for the problem under consideration, provides some preliminary insight into the type of calculations that must be performed in a finite element subroutine. Some preliminary results are required.

Change of Variables Formula (One-Dimensional Version)

Let $f: [x_1, x_2] \rightarrow \mathbb{R}$ be an integrable function and let $x: [\xi_1, \xi_2] \rightarrow [x_1, x_2]$ be continuously differentiable, with $x(\xi_1) = x_1$ and $x(\xi_2) = x_2$. Then

$$\int_{x_1}^{x_2} f(x) dx = \int_{\xi_1}^{\xi_2} f(x(\xi)) x_{,\xi}(\xi) d\xi \quad (1.15.1)$$

Chain Rule

Let f and x be as above, and, in addition, assume f is differentiable. Then

$$\frac{\partial}{\partial \xi} f(x(\xi)) = f_{,x}(x(\xi)) x_{,\xi}(\xi) \quad (1.15.2)$$

Proofs of these results may be found in [11].

The computation of k^e proceeds as follows:

$$\begin{aligned} k_{ab}^e &= \int_{\Omega^e} N_{a,x}(x) N_{b,x}(x) dx \quad (\text{by definition}) \\ &= \int_{-1}^{+1} N_{a,x}(x(\xi)) N_{b,x}(x(\xi)) x_{,\xi}(\xi) d\xi \\ &\quad (\text{Change of variables, where } x(\xi) \text{ is defined by (1.12.6)}) \\ &= \int_{-1}^{+1} N_{a,\xi}(\xi) N_{b,\xi}(\xi) (x_{,\xi}(\xi))^{-1} d\xi \end{aligned}$$

$$\begin{aligned}
 & \text{(Chain rule; } N_{a,\xi}(\xi) = (\partial/\partial\xi)N_a(x(\xi)) = N_{a,x}(x(\xi))x_{,\xi}(\xi)) \\
 & = (-1)^{a+b}/h^e \quad \text{(by (1.12.7)–(1.12.9))}
 \end{aligned}$$

Thus

$$\boxed{k^e = \frac{1}{h^e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}} \quad (1.15.3)$$

Observe that $N_{a,\xi}$ (see (1.12.7)) does not depend upon the particular element data, as $N_a = N_a(\xi)$. We shall see that this is generally true, and hence these computations may be done once and for all.

The derivatives $x_{,\xi}$ and $\xi_{,x}$ do depend on the particular element data (in the present case h^e), and subroutines will be necessary to compute the analogs of these quantities in more general cases.

Now we wish to compute f^e . However, this cannot be done without explicitly knowing what $\ell = \ell(x)$ is. In practice, it would be inconvenient to reprogram every time we wanted to solve a problem involving a different function ℓ . Generally a convenient approximation is made. For example, we might replace ℓ by its linear interpolate over each element, namely,

$$\ell^h = \sum_{a=1}^2 \ell_a N_a \quad (1.15.4)$$

where $\ell_a = \ell(x(\xi_a))$; see Fig. 1.15.1. The notation ℓ^h is used to indicate that the approximation depends upon the mesh. This represents an approximation that is sufficient for most practical applications. (It is, of course, exact for constant or linear “loading” of the element.) Now standardization of input to the program may be facilitated; that is, the nodal values of ℓ are the required data. Let us employ this approximation in the explicit calculation of an element force vector:

$$\begin{aligned}
 \int_{\Omega^e} N_a(x) \ell^h(x) dx &= \int_{-1}^{+1} N_a(x(\xi)) \ell^h(x(\xi)) x_{,\xi}(\xi) d\xi \quad \text{(change of variables)} \\
 &= \frac{h^e}{2} \sum_{b=1}^2 \int_{-1}^{+1} N_a(\xi) N_b(\xi) d\xi \ell_b \quad \text{(by (1.12.8))} \quad (1.15.5)
 \end{aligned}$$

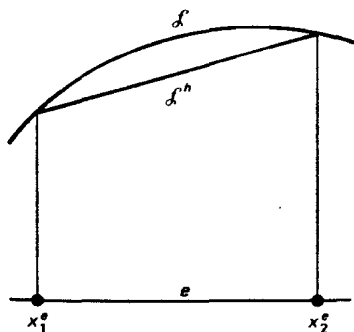


Figure 1.15.1 Approximation of ℓ by piecewise linear interpolation of nodal values.

Carrying out the integrations ($\int_{-1}^1 N_a N_b d\xi = (1 + \delta_{ab})/3$) yields

$$\begin{aligned} f^e &= \frac{h^e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix} \quad (+ \text{ boundary terms,} \\ &\quad \text{cf. (1.13.12)}) \\ &= \frac{h^e}{6} \begin{Bmatrix} 2f_1 + f_2 \\ f_1 + 2f_2 \end{Bmatrix} \quad (+ \text{ boundary terms}) \end{aligned} \quad (1.15.6)$$

Remark. It can be shown that, under suitable hypotheses, piecewise linear nodal interpolation produces $O(h^2)$ errors in the data; in this case, f . (See [12], pp. 56–57, for basic estimates of interpolation errors.) It can be shown that, in appropriate measures of the error, this produces at worst $O(h^2)$ errors in u^h and $u^h_{,x}$.

The following exercise indicates that there may be better ways to approximate given data.

Exercise 1. Suppose $f(x)$ is quadratic (i.e., consists of a linear combination of the monomials $1, x, x^2$). Determine a piecewise linear approximation—not necessarily continuous—to f over each element which results in exact nodal values. *Hint:* The analysis may be performed with respect to one element.

Exercise 2. The equation of a string on an elastic foundation is given by:

$$u_{,xx} - \lambda u + f = 0 \quad \text{on } \Omega =]0, 1[$$

where λ , a positive constant, is a measure of the foundation stiffness. Assuming the same boundary conditions as for the problem discussed previously in this chapter, it can be shown that an equivalent weak formulation is:

$$\int_{\Omega} (w_{,x} u_{,x} + w \lambda u) dx = \int_{\Omega} w f dx + w(0)h$$

where $u \in \mathcal{S}$, $w \in \mathcal{U}$, and so on. This can also be written as

$$a(w, u) + (w, \lambda u) = (w, f) + w(0)h$$

i. Let $u^h = v^h + q^h$. Write the Galerkin counterpart of the weak formulation:

$$\begin{aligned} a(w^h, v^h) + \boxed{} &= \\ (w^h, f) + w^h(0)h - a(w^h, q^h) & \\ - \boxed{} & \end{aligned}$$